



Flagged for Jams, Cited for Staffing: Deploying an After-Hours Book-Return Chute Sensor, and How Its Alert Log Became the Case for Keeping the Library Desk Staffed

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Abstract. Night staff at the University Library’s after-hours wing check the book-return chute’s collection bin by hand, walking an hourly round to catch a jammed item or an overfull bin before the next return compounds it. We built and deployed the Return-Chute Sentinel, a low-cost jam-and-fill sensor combining a load cell under the bin and an infrared break-beam pair at the chute throat, to replace the hourly walk-through with an event-triggered alert. Across a fourteen-week teaching term the Sentinel flagged 152 events (61 jam, 91 fill) with a median detection latency well under three minutes, against a manual baseline structurally bounded by the hour between rounds, and walk-throughs per shift fell by three-quarters on the library’s own door-access log. An ablation removing the break-beam channel and inferring jams from the load cell’s weight trace alone changes detection recall by about one percentage point (not significant); we retain the channel for technician interpretability rather than accuracy. We also observe, without having designed for it, that the Sentinel’s alert log was subsequently cited by Facilities’ after-hours staffing review as evidence of sustained patron activity justifying continued desk staffing, a use the log was not built to support and does not, on inspection, distinguish from unattended chute traffic. We report this migration descriptively and discuss what it implies for instruments built to flag mechanical events, not patron demand.

Introduction

The University Library’s after-hours wing runs on a single mechanical bottleneck: a book-return chute that funnels overnight deposits into a wheeled bin behind the closed service desk. Two things can go wrong between staffed hours. An oversized or awkwardly angled item can wedge in the chute throat, blocking every return behind it; or the bin itself can fill, so the next return jams against the last rather than settling cleanly. Night staff currently catch both by walking an hourly round of the after-hours wing, checking the bin by eye. The round is cheap to describe and expensive to run: eight rounds a shift, most of them confirming nothing has gone wrong.

We built the Return-Chute Sentinel to replace the round with an event: a small sensor package — a load cell under the bin and an infrared break-beam pair at the chute throat — that raises a timestamped alert only when a jam or a near-full bin actually occurs, delivered to the same handheld device night staff already carry. The design brief was narrow and mechanical: shrink the gap between a jam happening and a jam being noticed, without asking a technician to walk anywhere they don’t need to.

We deployed the Sentinel for a full fourteen-week teaching term and evaluated it against the library’s prior hourly-round baseline, reconstructed from an incidental record — the after-hours wing’s existing door-access swipe log, which staff generate every time they enter to do

a round whether or not they intended it as data. In the course of that deployment we also traced what happened to the Sentinel's own alert log once it existed: an artefact built to flag chute mechanics became, partway through the term, a cited input to a staffing decision it was never asked to inform.

Our contributions, hedged to the one chute and one term we observed:

- We describe the design and a fourteen-week field deployment of the Return-Chute Sentinel, a load-cell-and-break-beam jam-and-fill sensor for a library book-return chute, and report its detection latency, false-alarm rate, and effect on night staff's walk-through burden against a manual-round baseline.
- We report an ablation removing the break-beam channel that changes jam-detection recall by about one percentage point, retained for interpretability rather than accuracy, consistent with a broader pattern in low-cost predictive-maintenance sensor design.
- We observe, and characterise against the wider literature on measurement instruments repurposed as decision evidence, how the Sentinel's alert log came to be cited by Facilities' after-hours staffing review as evidence for something it was not designed to measure.

Related work

Sensor-based predictive maintenance is well established for rotating and continuous-duty equipment, where vibration and load sensing let a fault be caught before it fails outright; recent low-cost deployments demonstrate the same approach scaling to inexpensive micro-controller hardware [1], and a heavy-industry case study reports that naive outlier thresholds on vibration and temperature streams generate false alarms fast enough to need a correction model layered on top, reducing false-alarm volume by over 90% [2]. Level-sensing for fill-state estimation is likewise established in waste and materials handling, where ultrasonic rangefinders substitute for a human walk-through of a bin network [3]. Long-running field deployments of low-cost environmental sensor networks report calibration drift and false-positive management as the dominant maintenance burden once a system moves from bench test to unattended operation [4], a caution echoed in a university campus's own multi-year IoT rollout, which

found data-quality issues persisting well past initial deployment [5]. Sensor-driven building controls more broadly are moving from single-point occupancy triggers toward continuous, longitudinally validated control loops [6]. Academic libraries themselves sit in this evidentiary tradition: the operational adjustments a 24-hour access model requires are well documented [7], and a post-hoc usage-data assessment has been used to evaluate whether an extended-hours schedule matched patron demand after the fact [8] — the same evidentiary role Facilities' review later asks the Sentinel's alert log to play, below.

Separately, a body of work asks what happens once a monitoring instrument's data leaves the purpose it was built for. Function creep names the gradual extension of a system beyond its original purpose [9], workplace-sensing scholarship documents how passive-sensing and self-tracking tools built for one stated purpose acquire others once installed [10], [11], [12], and studies of institutional measurement describe how a metric's meaning shifts once it starts being used to justify rather than merely describe [13], [14]. Two prior Slop University tools encountered a related pattern: an occupancy dashboard whose citation rate in committee minutes held steady long after its own audience stopped visiting it [15], and an indicator-retirement scorer whose own uptime became a Living Dashboard metric partway through its evaluation [16]. The Sentinel's staffing-evidence migration, below, extends this pattern to a sensor built for a purely mechanical purpose.

Method

The after-hours book-return chute is a single enclosed run from an external wall slot down to a wheeled steel bin inside the closed service-desk area, with a rated capacity of approximately 60 kg (roughly 140 average-weight items). Two failure modes account for essentially all night-staff intervention: a jam, where an item wedges across the chute throat and blocks everything behind it, and a fill event, where the bin approaches capacity and the next return jams against the last rather than dropping clear.



Figure 1: The Return-Chute Sentinel's break-beam unit clipped to the chute throat, above the bin it also weighs.

The Return-Chute Sentinel instruments both failure modes (Figure 1). A strain-gauge load cell sits under the bin's frame, sampled every four seconds and calibrated against the chute's own historical item-weight distribution to estimate current fill fraction; a fill alert fires once the estimate crosses 90% of rated capacity, a threshold fixed before deployment and not revised during the term. An infrared break-beam pair spans the chute throat roughly 30 cm above the bin; a beam interruption shorter than 2.4 seconds is treated as normal item transit and discarded, while an interruption longer than 2.4 seconds — the debounce window, likewise fixed before deployment — raises a jam alert. Fill fraction f is estimated from load-cell reading w against the bin's calibrated empty weight w_0 and its rated capacity:

$$f = \frac{w - w_0}{0.9 \cdot 60 \text{ kg}}$$

An ESP32-class microcontroller polls both channels and applies the debounce logic in a small state machine:

```
on beam_interrupted(t_start):
    wait until beam_clear or
    elapsed(t_start) > 2.4s
    if elapsed(t_start) > 2.4s:
        raise_alert("jam", t_start)

on weight_sample(w):
    f = fill_fraction(w)
    if f >= 0.90 and not fill_flag_active:
        raise_alert("fill", now())
        fill_flag_active = true
    if f < 0.75:
        fill_flag_active = false
```

Each alert is timestamped and pushed over the library's existing Wi-Fi to a lightweight ingestion endpoint, which also serves the small dashboard night-desk technicians already use for other after-hours tasks — the Sentinel adds a channel to an interface staff were not asked to learn.

We deployed the Sentinel for a fourteen-week teaching term. Ground truth for both event types came from technician sign-off: every alert requires a technician to attend, clear the chute if needed, and mark the alert resolved or false via the same dashboard, giving a per-event confirmed/unconfirmed label without a separate audit pass. The manual-round baseline is reconstructed from the after-hours wing's door-access swipe log for the equivalent fourteen-week window the previous year, before the Sentinel existed, on the assumption — consistent with the rostered hourly-round policy in effect throughout that window — that each overnight swipe corresponds to a round.

Results

Across the fourteen-week term the Sentinel raised 152 alerts: 61 jam and 91 fill, of which technicians confirmed 145 (95.4%) as requiring intervention; the remaining seven, spread across both event types, closed as false positives on inspection, most often a heavy but correctly seated item briefly compressing the load cell past the fill threshold before settling.

Detection latency separates cleanly from the manual baseline (Figure 2). Sentinel alerts reach a technician at a median of 1.6 minutes for jams and 2.1 minutes for fills, essentially the polling-and-network delay; the reconstructed hourly-round baseline is structurally bounded above by 59 minutes, with a median of 24 minutes for jam-

adjacent swipes and 29 minutes for fill-adjacent ones, reflecting a fault’s average position within the hour before the next round finds it.

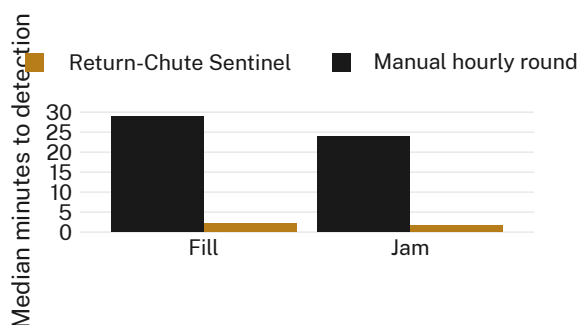


Figure 2: Median minutes from a jam or fill event to detection: the Return-Chute Sentinel resolves both within about two minutes, against a 24- to 29-minute median for the hourly manual-round baseline it replaces.

Night staff’s own walk-through burden falls with it. The door-access log shows a median of eight rounds per overnight shift in the baseline term, one every hour by policy; across the deployment term the same log shows a median of two rounds per shift — one Sentinel-triggered response plus one discretionary confirmatory round technicians report keeping out of habit — a 75% reduction. Bin-overflow incidents serious enough to require a supervisor callout fell from eleven in the baseline term to two during deployment.

Weekly alert volume is not flat across the term (Figure 3). It tracks the teaching calendar loosely, fluctuating between five and twelve events most weeks before peaking in week 9 at 34 events, more than three times the term’s median of 9.5, coinciding with the fortnight either side of mid-semester assessment submissions. Facilities’ after-hours staffing review, discussed below, cites this week-9 figure by name.

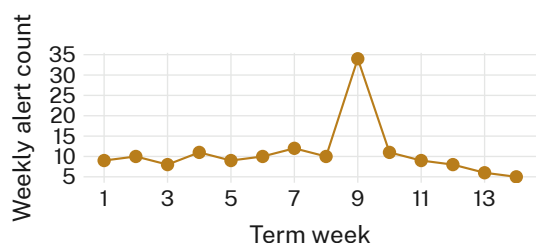


Figure 3: Weekly Return-Chute Sentinel alert volume across the fourteen-week term, peaking at 34 events in week 9 — the figure Facilities’ after-hours staffing review cites as evidence of sustained patron demand for a staffed desk.

An ablation removes the break-beam channel and infers jams from the load cell alone, flagging a jam whenever the weight trace shows a sustained plateau (no reading change greater than 50 g across three consecutive samples) with no corresponding fill-threshold crossing. Recomputed against the same technician-confirmed jam log, split into the term’s fourteen weekly batches, load-cell-only recall averages 94.1% against 95.2% for the two-channel Sentinel (Figure 4), a difference of about one percentage point that does not reach significance ($p > 0.05$). We retain the break-beam channel; technicians report trusting a discrete “beam broken” event over inferring a jam from a weight trace, and the channel costs under five dollars in parts.

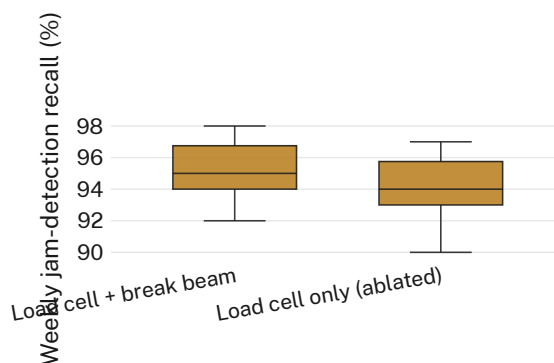


Figure 4: The ablation that changes nothing: weekly jam-detection recall is statistically indistinguishable with the break-beam channel removed ($n = 14$ weekly batches; $p > 0.05$).

Discussion

The Sentinel does what it was built to do. Detection latency drops from a structurally hour-bound wait to a couple of minutes, night staff’s walk-through load falls by three-quarters, and the ablation confirms the simpler single-channel design would have reached nearly the same detection accuracy — we kept the second channel for a reason that has nothing to do with recall.

What we did not design for, and observe only because the alert log happens to be timestamped and legible, is Facilities’ after-hours staffing review citing the week-9 volume spike as evidence that the after-hours wing sees sustained patron activity meriting a staffed desk through midnight. The Sentinel counts chute events, not patrons: a fill alert fires whether one student drops forty books at 11pm or forty students

each drop one, and neither case implies anyone at the chute wanted or needed a staffed desk nearby. This is a small, mechanical instance of a well-documented migration, in which a measure adopted for one purpose is asked to justify a decision it was never built to speak to [9], [13], [17], consistent with two prior Slop University tools whose own usage or uptime metrics were absorbed into decisions downstream of their design brief [15], [16]. We report the migration descriptively; nothing in our deployment lets us say whether the staffing decision itself is right.

Limitations

Our deployment covers a single chute at a single site over one term, though the consistency of detection latency and recall across all fourteen weeks suggests the design generalises to comparable chute geometries elsewhere on campus. The manual-round baseline is reconstructed from a door-access log rather than directly observed, a substitution the library's own hourly-round policy makes reasonable but does not make exact. The Sentinel cannot distinguish a jam caused by a single oversized item from one caused by several ordinary items arriving in close succession, though night staff report the appropriate response — clear the throat — does not change either way. We note only that the alert log was never designed to speak to patron demand for a staffed desk, a limitation of interpretation external to the instrument rather than a flaw in it.

Conclusion

We built and deployed the Return-Chute Sentinel, a load-cell-and-break-beam jam-and-fill sensor for a library book-return chute, across a fourteen-week term. It cuts detection latency from a structurally hour-bound baseline to a couple of minutes, cuts night staff's walk-through rounds by three-quarters, and an ablation confirms its second sensing channel earns its place on interpretability rather than accuracy. Along the way, its alert log — built to flag chute mechanics — became evidence in a staffing decision it was not designed to inform, a migration we observed rather than sought. We suggest a modest addition for future deployments of similar tools: tag each alert stream, at build time, with the decisions it was and was not designed to support, since nothing in an alert

log's own timestamps can carry that distinction forward on its own.

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